CS 301: COMPUTER SYSTEMS & OPERATING SYSTEMS

Lecture 04: Process vs. Thread Management

Instructor: Prof. A. Vance **Term:** Fall Semester **Category:** Core Systems

1. Key Architectural Concepts

Process

An isolated instance of a program in execution. Contains its own memory address space, program counter, stack, heap, and resource handles managed explicitly by the OS kernel.

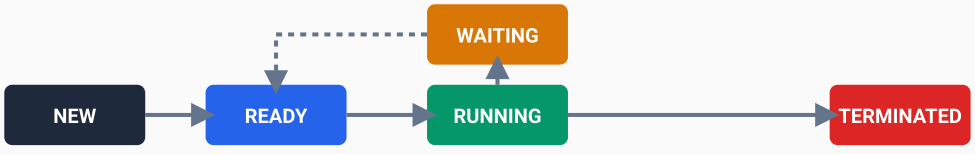
Thread

The smallest dispatchable unit of execution within a process. Multiple threads share the parent process's address space, code segment, and global data while maintaining private stack registers.

2. Feature Comparison Matrix

Feature	Process	Thread
Memory Context	Isolated & Protected Address Space	Shared within parent Process space
Creation Overhead	High (Syscall + Page table allocation)	Low (Lightweight execution context)
Context Switch	Slow (TLB flush & CPU cache miss)	Fast (Registers swap, memory intact)
Fault Isolation	High (Process crash is contained)	Low (Thread fault crashes process)

3. CPU Process Lifecycle & Transitions



Exam Tip: Remember that Kernel Threads are scheduled directly by the OS, whereas User Threads are managed entirely by a user-space thread library without kernel context switches.